



Master Presenter's Manual

For

Software Testing using Selenium

Curriculum Code: OV-1516

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Ver 1.0*

Amendment Record

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1. Introduction

At the end of this module, student will be able to:

- Learn Test Automation
- Understand Selenium Test Suite of tools
- Outline the use of Java for Selenium
- Learn Web Locators and Web Elements
- Define Selenium Scripts in Eclipse IDE for Test Execution
- Learn Database Testing using Java and Selenium
- Understand Selenium IDE and Web Driver
- Understand Data Driven, Keyword Driven and Hybrid Selenium Test Framework and Selenium Grid.

2. Information on Session Allocation

Module	Online (No. of Hours)	Skill Code
STSE	16	1239

STSE: Software Testing using Selenium

3. Module Deliverables

The deliverables to be provided to the students and the faculty have been tabulated below:

➤ **Deliverables for Students:**

Module	Deliverable
Software Testing using Selenium	Kit Code: OV-1516-KIT01-01-A Kit Name: Kit of 1 title for OV-1516 SW Testing with Selenium Book Code: AINNSELEN10319E000 Book Name: Software Testing using Selenium

➤ **Deliverables for Faculty:**

Module	Deliverable
Software Testing using Selenium	Kit Code: OV-1516-KIT01-01-A Kit Name: Kit of 1 title for OV-1516 SW Testing with Selenium Book Code: AINNSELEN10319E000 Book Name: Software Testing using Selenium
Course	Master PM Classroom Presentations Trainer Guide

To aid the teaching process, following are the deliverables

Student Deliverables:

1. Learner's Guide (eBook)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Show Me How - Student can view a step-wise simulation/demonstration of the module related topics.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Week-wise Session Schedule

- A Session is of 2 hours duration

Week	Day 1	Day 2	Day 3	Day 4
1	Session 1 STSE – TL1	Session 2 STSE – TL2	Session 3 STSE – TL3	Session 4 STSE – TL4
2	Session 5 STSE – TL5	Session 6 STSE – TL6	Session 7 STSE – TL7	Session 8 STSE – TL8

TL – Online Session

5. Session Coverage

The table below details out the contents of each session where:

Session No.	Session Title	Session Details	Deliverables' Mapping
1	STSE – TL1	All the topics as listed below from Session 1 and Session 2 of <i>Software Testing using Selenium</i> book should be covered in this session. <u>Session 1 – Introduction to Selenium</u> ➤ Explain Selenium ➤ Describe benefits of automation testing ➤ Explain advantages of Selenium ➤ Differentiate between Selenium and Quick Test Professional (QTP) ➤ List the steps involved in setting initial configuration for using Selenium <u>Session 2 – Selenium Test Suite</u> ➤ Identify components of the Selenium suite ➤ Describe Selenium IDE ➤ Describe Selenium RC and its architecture ➤ Describe Selenium WebDriver ➤ Explain working of the Selenium Grid ➤ Differentiate between Selenium IDE, RC, and WebDriver	Software Testing using Selenium SG – Session 1 & 2 XP – Session 1 & 2 TG – Session 1 & 2
2	STSE – TL2	All the Workshop topics from Session 1 and Session 2 of <i>Software Testing using Selenium</i> course should be covered in this session.	Software Testing using Selenium (Onlinevarsity) SG – Session 1 & 2
3	STSE – TL3	All the topics as listed below from Session 3 of <i>Software Testing using Selenium</i> book should be covered in this session. <u>Session 3 – Web Locators and</u>	Software Testing using Selenium SG – Session 3 XP – Session 3 TG – Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Selenium Commands</u> ➤ Identify different types of Web locators ➤ Describe Selenium functions for handling user interactions ➤ Compare assertion and verification ➤ Identify different types of Selenium commands ➤ Explain the significance of different wait commands	
4	STSE – TL4	All the topics as listed below from Session 4 of <i>Software Testing using Selenium</i> book should be covered in this session. <u>Session 4 – Selenium WebDriver and Java</u> ➤ Describe the Java programming language ➤ Explain the object-oriented concepts ➤ Explain how to set up the Java environment ➤ Identify the data types, variable types, and operators in Java ➤ Describe the various control flow statements in Java ➤ Explain the significance of exception handling ➤ Explain the test script in Selenium RC ➤ Explain the configuration of Selenium WebDriver in Eclipse ➤ Explain the debugging in Eclipse	Software Testing using Selenium SG – Session 4 XP – Session 4 TG – Session 4
5	STSE – TL5	All the Workshop topics from Session 3 and Session 4 of <i>Software Testing using Selenium</i> course should be covered in this session.	Software Testing using Selenium (Onlinevarsity) SG – Session 3 & 4
6	STSE – TL6	All the topics as listed below from Session 5 of <i>Software Testing using Selenium</i> book should be covered in this session.	Software Testing using Selenium SG – Session 5

Session No.	Session Title	Session Details	Deliverables' Mapping
		<u>Session 5 – Selenium Tests and Frameworks</u> ➤ Describe the benefits of frameworks ➤ Identify the types of frameworks ➤ Explain the significance of TestNG framework in Selenium ➤ Describe the features of TestNG framework ➤ Explain how to execute scripts using the TestNG framework	XP – Session 5 TG – Session 5
7	STSE – TL7	All the topics as listed below from Session 6 of <i>Software Testing using Selenium</i> book should be covered in this session. <u>Session 6 – Grid and Test Design Techniques</u> ➤ Describe the significance of Selenium Grid ➤ Explain the configuration of Selenium Grid components ➤ List the advantages of Selenium Grid ➤ Explain the significance of the Apache Poor Obfuscation Implementation (POI) API and its components ➤ Explain the significance of the Log4j API and its components ➤ Describe the significance of the Java Database Connectivity (JDBC) API	Software Testing using Selenium SG – Session 6 XP – Session 6 TG – Session 6
8	STSE – TL8	All the Workshop topics from Session 5 and Session 6 of <i>Software Testing using Selenium</i> course should be covered in this session.	Software Testing using Selenium (Onlinevarsity) SG – Session 5 & 6

6. Library References

➤ Selenium Testing Tools Cookbook by Unmesh Gundecha
➤ Mastering Selenium Webdriver by Mark Collin
➤ Test Automation using Selenium WebDriver with Java: Step by Step Guide by Navneesh Garg

7. Evaluation Strategy

The Evaluation Strategy for this course is as follows:

- Final Examination

7.1 Award of Grades

Overall Weighted Marks $\geq 40\%$ but $< 60\%$ qualifies for **PASS**
Overall Weighted Marks $\geq 60\%$ but $< 75\%$ qualifies for **CREDIT and**
Overall Weighted Marks $\geq 75\%$ qualifies for **DISTINCTION**

Note: To attain a PASS/CREDIT/DISTINCTION grade, a student should achieve at least 40% in Final Examination, otherwise he/she will be declared as 'Referred'.

4.2 Final Examination (External)

Below is the list of modules to be covered in Final Examination along with their weightages:

Final Examination 1 (E1)

Marks: 50
Duration: 1 hour
Type: Objective

S#	Name of the Subject	Marks
1.	Software Testing using Selenium	50

Final Examination 2 (R1)

Marks: 50
Duration: 2 hours
Type: Practical

S#	Name of the Subject	Marks
1.	Software Testing using Selenium	50

Evaluation Details

Practical – 50%
OES – 50%

8. Classroom and Lab Pre-requisites

The requirements for conducting the sessions have been described as follows:

8.1 Software Requirements

Microsoft Windows 11
Java SE 7 or higher
Selenium 2.9.1.1-signed

8.2 Recommended Hardware

Intel Core i3/i5 Processor or higher
8 GB RAM or above
Color SVGA
500 GB Hard Disk space
Mouse
Keyboard

8.3 Other Requirements (For Classroom)

White board with marker
Computer for giving demonstrations wherever applicable
Computer / LCD projector for presenting the transparencies

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